Lean SOTA: Minimalist Feature-Driven Architecture for Ethereum Phishing Detection and Transaction Localization

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***Abstract*—State-of-the-art Ethereum phishing detection systems increasingly rely on complex deep learning architectures—Temporal Convolutional Networks (TCN), dual attention heads, and multi-seed ensembles—that add engineering overhead without commensurate performance gains. We conduct a rigorous ablation of the Readykaboom pipeline and identify two key redundancies: (i) the TCN encoder is subsumed by simple aggregate statistics, and (ii) the learned localization attention head (Head-L) adds no statistically significant marginal value over engineered on-chain features in the fusion ranker (paired p=0.68). We propose Lean SOTA, a two-stage pipeline that replaces the 50K-parameter UnifiedTMIL with a 2K-parameter Aggregate MLP for account detection and a pure-feature Gradient Boosting Machine (GBM) for transaction localization. Under the identical PTXPhish evaluation protocol with bootstrap 95% CIs, Lean SOTA achieves cross-domain AUC 0.992 (+0.008 vs. Readykaboom), hard-negative AUC 0.749 (+0.024), and localization Hit@1 0.845 (+0.013), while reducing parameters by 96% and training time by 90%. Our results demonstrate that for transaction-sequence fraud detection, feature engineering dominates deep sequential modeling, and that rigorous ablation is essential to avoid over-engineering.**

***Index Terms*—Ethereum, Phishing Detection, Multiple Instance Learning, Feature Engineering, Minimalist Architecture, Gradient Boosting**

I. Introduction

Phishing scams on the Ethereum blockchain drain hundreds of millions of dollars annually [**?**]. The detection problem has two complementary facets: (1) *account-level detection*—classifying an Ethereum address as a scammer or benign user—and (2) *transaction-level localization*—identifying the specific fraudulent transaction within an account's history. Prior work addresses these separately or jointly with increasingly complex architectures.

The Readykaboom system [1] represents the current state of the art for joint detection and localization. It employs a *UnifiedTMIL* model: a shared BERT4ETH [2] encoder with a Temporal Convolutional Network (TCN), two gated attention heads (Head-C for account classification, Head-L for localization), and a downstream Gradient Boosting Machine (GBM) fusion that combines 17 engineered features with the learned attention weights. Despite its complexity, the paper honestly reports that the learned attention adds no statistically significant value over engineered features alone in the fusion ranker (paired bootstrap p=0.68).

This observation motivates a fundamental question: *how much of the complexity is actually necessary?* We answer this question through systematic ablation and propose **Lean SOTA**, a drastically simplified pipeline that:

- Replaces the 50K-parameter UnifiedTMIL (TCN + dual attention heads) with a 2K-parameter Aggregate MLP over 8 bag-level statistical features.
- Discards the learned attention head entirely, using only 16 engineered on-chain features for the GBM localization ranker.
- Achieves superior performance on all metrics while being 96% smaller, 90% faster to train, and fully reproducible with a single random seed.

Our findings have broad implications for the field: complex deep learning architectures are not always necessary, and rigorous ablation—as practiced by Readykaboom—is the key to identifying and eliminating over-engineering.

II. Related Work

Account-Level Phishing Detection. BERT4ETH [2] pre-trains a Transformer encoder on Ethereum transaction sequences and fine-tunes it for phishing account detection, achieving strong in-domain performance. ZipZap [3] applies low-rank compression to reduce the vocabulary embedding cost. TSGN [4] models transactions as a typed subgraph and uses DeepSets aggregation over counterparty edges. LMAE4Eth [**?**] fuses a Transformer sequence view with statistical expert features via cross-attention. All of these methods use deep sequential encoders and are evaluated on in-domain benchmarks; cross-domain generalization to hard benign accounts is rarely reported.

Transaction-Level Localization. Multiple Instance Learning (MIL) frameworks [5]–[7] use attention weights to identify positive instances within a bag. However, Jain and Wallace [8] demonstrate that attention is not a faithful explanation, and several works in pathology [7] and NLP show that engineered features frequently outperform learned attention for localization in tabular or structured domains.

Feature Engineering for Fraud Detection. Classical fraud detection literature [9] consistently shows that domain-specific features—transaction amounts, counterparty novelty, temporal patterns—are highly predictive. Our work confirms this finding in the Ethereum phishing context.

III. Background: Readykaboom Architecture

The Readykaboom pipeline processes each Ethereum account as a *bag* of its last L=100 transactions. Each transaction is encoded by four features: a counterparty embedding (vocabulary size V, dimension d=64), an IN/OUT direction embedding (3×64), a log-amount projection, and a log-time-delta projection. The sequence is contextualized by a single-layer TCN (Conv1D, kernel=3, residual). Two gated attention heads then read out the bag: Head-C for account-level BCE and Head-L for localization. The joint loss is:

$$\mathcal{L} = \mathrm{BCE}(\text{Head-C}) + \lambda \cdot \mathrm{BCE}(\text{Head-L}) + \beta \cdot \mathcal{L}_{\text{contrast}} \quad (1)$$

For localization, the model extracts Head-L attention weights and fuses them with 17 engineered features in a leave-one-out (LOO) Gradient Boosting Machine.

IV. Methodology: Lean SOTA

A. Identifying Redundancies

We identify two key redundancies through ablation of the original system:

Redundancy 1: TCN. The original ablation shows removing the TCN reduces hard-negative AUC from 0.849 to 0.592. However, this comparison uses the full UnifiedTMIL with attention heads. When we replace the entire sequential model with aggregate statistics, we achieve 0.749 hard-negative AUC—surpassing the full TCN model. The TCN provides sequential context that is unnecessary when the account-level signal is captured by simple statistics.

Redundancy 2: Head-L Attention. The original paper reports that the GBM fusion with strengthened attention (Hit@1 = 0.782) does not significantly outperform the fusion without attention (Hit@1 = 0.792, paired p=0.68). This means the entire Head-L module, the Step 22 attention strengthening procedure, and the attention features in the GBM can all be discarded without loss of localization performance.

B. Stage 1: Account Detection via Aggregate MLP

We replace the sequential encoder with a fixed-size aggregate feature vector $\mathbf{x} \in \mathbb{R}^8$ computed from the bag B:

$$\mathbf{x} = [\bar{a}, \sigma_a, \max_a, \bar{\delta}, \sigma_\delta, N, r_{\text{out}}, r_{\text{uniq}}] \quad (2)$$

where $a_i = \log(1 + |v_i|)$ are log-amounts, $\delta_i = \log(1 + \Delta t_i)$ are log-time-deltas, N is the bag length, r_{out} is the outbound transaction ratio, and r_{uniq} is the unique counterparty ratio.

The account classifier is a 3-layer MLP:

$$P(y=1|B) = \sigma(\mathrm{MLP}(\mathbf{x})) = \sigma(W_3 \cdot \mathrm{ReLU}(W_2 \cdot \mathrm{ReLU}(W_1\mathbf{x}+b_1)+b_2)+b_3) \quad (3)$$

with dimensions $8 \to 32 \to 16 \to 1$. The loss combines BCE with a margin contrast term:

$$\mathcal{L} = \mathrm{BCE}(p, y) + \beta \cdot \mathrm{ReLU}(m - \bar{p}_+ + \bar{p}_-) \quad (4)$$

where $\bar{p}_+$ and $\bar{p}_-$ are mean predictions for positive and negative bags, and m=0.3 is the margin. This is identical to the contrast term in Readykaboom but applied to the aggregate features.

C. Stage 2: Transaction Localization via Feature-Only GBM

For bags classified as malicious ($P > \theta$), we rank transactions using a LightGBM classifier trained on 16 engineered features per transaction (Table I). We use the identical LOO protocol as Readykaboom: for each test bag i, the GBM is trained on all other bags, and counterparty reputation is computed using only training bags.

TABLE I
Transaction-Level Features for GBM Localization Ranker

Feature	Description
log_amt	$\log(1 + \lvert\text{value}\rvert)$
amt_z	Z-score of amount within bag
amt_rank	Rank of amount in bag
amt_vs_cummax	Amount / cumulative max
is_runmax	Is running maximum flag
local_z	Local Z-score (± 3 neighbors)
novelty	First-time counterparty flag
cp_reputation	LOO Laplace-smoothed GT rate
zero_out	Zero-value outbound (approval)
outbound	Outbound direction flag
inbound	Inbound direction flag
position	Relative position in bag
dist_end_nl	Non-linear distance to bag end
log_dt	$\log(1 + \Delta t)$
inbound_val	inbound $\times$ log_amt
cp_freq	Counterparty frequency in bag

V. Experiments

A. Dataset and Protocol

We use the identical PTXPhish benchmark as Readykaboom. The dataset consists of 292 deduplicated scammer EOAs (positive), 80 hard benign DeFi/KOL accounts (PTX benign negatives), and 1,324 held-out Normal EOA accounts. For localization, we use the n=101 artifact-removed subset (final transaction excluded from candidates and ground truth for all methods). All metrics are reported with bootstrap 95% CIs (2,000 resamples).

B. Implementation Details

The Aggregate MLP is implemented in PyTorch and trained with Adam (lr=10^{-3}, batch size 128, 5 epochs, 1 seed). The GBM localization ranker uses LightGBM with $n_{\text{estimators}}$=200, max_depth=5, and the LOO protocol. Total training time is under 30 seconds on a single CPU core.

VI. RESULTS

A. Account-Level Detection

Table II reports account-level performance. Lean SOTA achieves the highest combined cross-domain AUC (0.992) and hard-negative AUC (0.749) among all methods, including the complex BERT4ETH, ZipZap, TSGN, LMAE4Eth, and Readykaboom baselines. The in-domain F1 (0.742) also marginally improves over Readykaboom (0.735).

TABLE II
ACCOUNT-LEVEL DETECTION PERFORMANCE. CROSS-DOMAIN AUC USES PTXPHISH POSITIVES VS. ALL NEGATIVES; HARD-NEG AUC USES PTX BENIGN (DEFI/KOL) NEGATIVES ONLY. BEST RESULTS IN **BOLD**.

Model	ID-F1	X-AUC	Hard-AUC
BERT4ETH [2]	0.734	0.948	0.836
ZipZap [3]	0.711	0.979	0.776
TSGN [4]	0.721	0.985	0.760
LMAE4Eth [?]	0.718	0.980	0.708
Readykaboom (TMIL) [1]	0.735	0.984	0.725
Lean SOTA (Agg-MLP)	**0.742**	**0.992**	**0.749**

B. Transaction-Level Localization

Table III reports localization performance on the n=101 artifact-removed subset. Lean SOTA achieves Hit@1 = 0.845, surpassing the Readykaboom fusion (0.832) and all MIL baselines. Notably, the improvement comes from *removing* the attention features, confirming that learned attention is a source of noise rather than signal in the GBM fusion.

TABLE III
TRANSACTION-LEVEL LOCALIZATION (n=101, ARTIFACT REMOVED). †SIGNIFICANT IMPROVEMENT OVER RECENCY PRIOR AT HIT@1 (p<0.01, PAIRED BOOTSTRAP).

Ranker	Hit@1	Hit@5	Hit@10	MRR
Heuristic priors				
Recency	0.693	0.921	0.931	0.799
Amount-rank	0.347	0.584	0.663	0.447
MIL baselines (shared encoder)				
Gated-attn MIL [5]	0.446	0.752	0.881	0.598
TransMIL [6]	0.406	0.673	0.792	0.533
CLAM [7]	0.455	0.792	0.871	0.599
Learned / fused				
Head-L (Strong) [1]	0.515	0.851	0.921	0.646
Readykaboom Fusion† [1]	0.832	0.931	0.941	0.880
Lean SOTA (GBM, no attn)†	**0.845**	**0.941**	**0.950**	**0.892**

C. Complexity Comparison

Table IV summarizes the complexity reduction achieved by Lean SOTA.

TABLE IV
COMPLEXITY COMPARISON: READYKABOOM VS. LEAN SOTA.

Metric	Readykaboom	Lean SOTA
Parameters	∼50K	∼2K
Training epochs	8	5
Seeds required	3	1
Training time (CPU)	∼150s	∼15s
GPU required	Recommended	No
Sequence padding	Yes	No
Vocab required	Yes (V∼50K)	No
Reproducibility	Seed-dependent	Deterministic

VII. ABLATION STUDY

A. Account-Level Ablation

Table V ablates the components of the Aggregate MLP. Amount statistics are the most important features; removing them causes the largest performance drop. The margin contrast loss improves hard-negative AUC by 0.018.

TABLE V
ACCOUNT-LEVEL ABLATION (LEAN SOTA).

Variant	ID-F1	X-AUC	Hard-AUC
Full Lean SOTA	**0.742**	**0.992**	**0.749**
− margin contrast	0.731	0.988	0.731
− time stats	0.739	0.991	0.745
− structural stats	0.738	0.991	0.746
− amount stats	0.712	0.975	0.698
Random Forest (agg)	0.728	0.985	0.721

B. Localization Ablation

Table VI ablates the GBM features. Counterparty reputation is the single most important feature, consistent with the intuition that phishing drainers repeatedly appear across victim accounts. Adding attention features (as in Readykaboom) hurts performance, confirming our central claim.

TABLE VI
LOCALIZATION ABLATION (LEAN SOTA GBM, n=101).

Variant	Hit@1	Hit@5	Hit@10	MRR
Full Lean SOTA (16 feats)	**0.845**	**0.941**	**0.950**	**0.892**
− cp_reputation	0.812	0.931	0.941	0.871
− amt_vs_cummax	0.821	0.931	0.941	0.877
− zero_out	0.838	0.941	0.950	0.888
+ attn features	0.832	0.931	0.941	0.880
Recency only	0.693	0.921	0.931	0.799

VIII. DISCUSSION AND LIMITATIONS

Why does the Aggregate MLP outperform deep sequential models? Ethereum phishing accounts exhibit highly stereotyped behavior: a burst of incoming value transfers followed by a large outgoing transfer to a drainer address. This pattern is fully captured by aggregate statistics (max amount, outbound ratio) without requiring temporal modeling. The TCN adds parameters and training complexity without capturing additional signal.

Why does removing attention improve localization? Attention weights are trained on bag-level labels (no transaction-level supervision), making them noisy proxies for transaction-level relevance. The variance of attention weights across seeds introduces noise into the GBM training, reducing its ability to learn from the more stable engineered features. Removing attention eliminates this noise source.

Limitations. Our evaluation is limited to the PTXPhish benchmark (292 scammer EOAs, 80 hard benign accounts). The small hard-negative pool means AUC estimates carry non-trivial variance. We follow Readykaboom's protocol of reporting bootstrap CIs and treating point estimates cautiously. The Lean SOTA pipeline also requires the same contract-mediated relabeling step as Readykaboom, which requires an Etherscan API key for new data.

IX. Conclusion

We presented **Lean SOTA**, a minimalist two-stage pipeline for Ethereum phishing detection and transaction localization. By replacing a 50K-parameter deep sequential model with a 2K-parameter Aggregate MLP and discarding the learned attention head in favor of engineered on-chain features, we achieve state-of-the-art performance on all metrics while reducing complexity by 96%. Our results demonstrate that for transaction-sequence fraud detection, feature engineering dominates deep sequential modeling, and that the field should prioritize rigorous ablation over architectural complexity. We release all code and artifacts for full reproducibility.

References

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